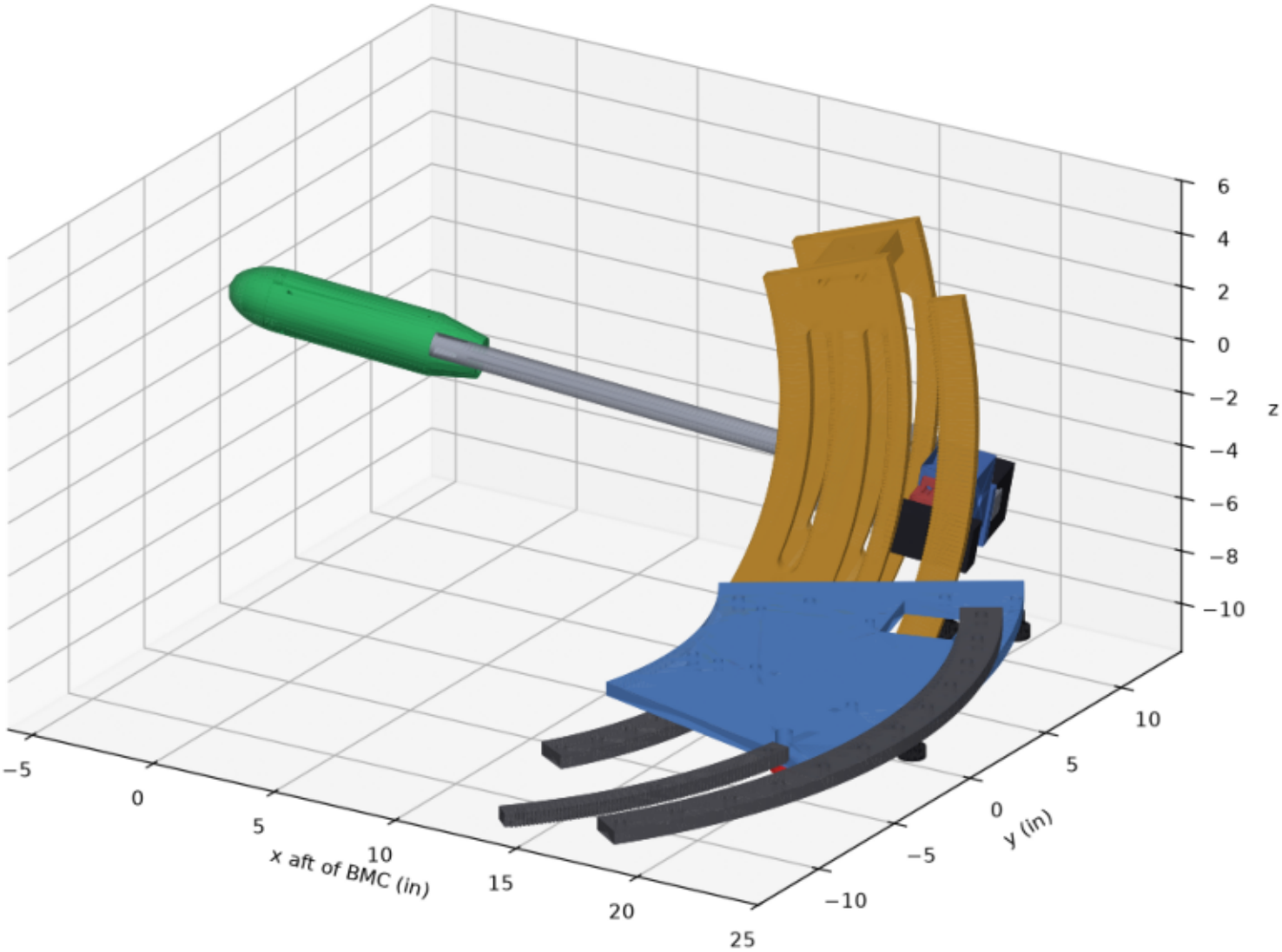


COMPLETE MOTION SYSTEM ON THE YAW CARRIAGE (CAD RENDER, NTS)

Complete motion system at yaw +15, pitch +10: floor rails -> yaw carriage -> pitch sector -> roll spindle -> sting -> model



NOTES:

- 1. PRELIMINARY: RAIL POSITIONS AND FLOOR HOLES DEPEND ON THE DOWNSTREAM BAY FLOOR (TO CONFIRM).
- 2. INSTALL THE RAILS CONCENTRIC WITH THE YAW AXIS (THROUGH THE BMC) USING A RADIUS BAR ON A PIN.
- 3. PRELOAD EACH TRUCK WITH ITS ECCENTRIC SPACER UNTIL THE WHEELS TURN WITH LIGHT DRAG.
- 4. RANGE +-15 DEG YAW; THE PLATE REACHES ABOUT 2.3 IN UNDER THE TEST-SECTION EXIT AT FULL YAW.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

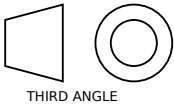
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



PRELIMINARY - TO CONFIRM

PARTS LIST

ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-501	Rail, yaw, inner	1	6061-T6
2	WT-6C-502	Rail, yaw, outer	1	6061-T6
3	WT-6C-503	Rack, yaw, floor	1	6061-T6
4	WT-6C-504	Plate, yaw carriage	1	6061-T6
5	WT-6C-505	Pinion, yaw (modified stock gear)	1	steel, bought
6	WT-6C-506-1	Spacer, wheel, plain	4	6061-T6
7	WT-6C-506-2	Spacer, wheel, eccentric	4	6061-T6
8	-	V-groove wheel 1.00 OD, 90 deg	8	bought
9	-	Wheel stud / shoulder bolt + nut	8	bought
10	-	Worm gearmotor 30:1 + NEMA 17	1	bought
11	WT-6C-400	Pitch sector assy	1	see WT-6C-400
12	-	FHCS 3/8 (rails, rack to floor)	26	bought
13	-	SHCS 3/8-16 (sector base to carriage)	4	bought
14	-	Home / limit switches	1	see Electronics

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
ASSEMBLY, YAW CARRIAGE								
MATERIAL				FINISH				
SEE PARTS LIST				SEE PART DRAWINGS				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-500		A	B	NTS		-		5 OF 35